Zain Rizvi

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EDUCATION

Honours Bachelor of Science, University of Toronto — Sep 2022 - Aug 2026
Computer Science & Mathematics Double Major
Relevant Coursework: Machine Learning, Artificial Intelligence, Probability, Data Structures, Algorithms
Awards: 1st Place at AMACSS AI Olympics Hackathon, Frank Faubert Scholarship, Admission Scholarship

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, R, C
Libraries: PyTorch, Scikit-learn, XGBoost, LangChain, Spring Boot, Pandas, NumPy, Polars
Tools: Docker, AWS (Cloud Practitioner), Git/GitHub, REST APIs, Postman, MySQL, Cloudflare

WORK EXPERIENCE

Data Scientist (Co-op) — May 2025 - Dec 2025
Purolator Inc. — *Mississauga, ON*

- Cut national operations forecast error by **75%** (4% to 1% MAPE) vs. previous biweekly forecast, cross-validated on 5 years of historical data across **10M+ shipments**
- Enabled route-level staffing and capacity decisions (**~50x more forecasts**) by compressing 50+ routes into **5 PCA components** and forecasting each with **STL + ARIMA** model
- Automated end-to-end freight forecasting pipelines with Python and R, reducing runtime from **25 min to 3 min** and extending support for new terminals via historical demand reconstruction
- Gave **100+ stakeholders** real-time visibility into capacity planning and network optimization by automating executive KPI reporting with Python, QlikView, and Excel

Data Analyst (Co-op) — May 2024 - Aug 2024
Ontario Ministry of Transportation — *Toronto, ON*

- Reconciled **$1M+** across **130,000+ records** for Heavy Commercial Vehicle payments with algorithmic outlier detection in Python and Excel; automation cut processing time by **95%**
- Developed and maintained Spring Boot REST APIs for the Towing and Storage Safety and Enforcement Act Application, ensuring reliability and AODA accessibility compliance

PROJECTS

Robot Learning with Vision-Language-Action Models — robotics.zainrizvi.ca

- Fine-tuned the action expert of **SmolVLA** (450M-parameter vision-language-action model) to follow **natural-language commands**, reaching 9/10 grasp and 5/10 stack success
- Visualized attention across all 16 SmolVLA layers in an interactive **React/Three.js** web app to support **mechanistic interpretability**
- Built a 7-joint arm simulation (**MuJoCo, Gymnasium, PyTorch**) with scripted experts reaching **87%** on bin throws and **90%** on peg insertions

Rocket League Replay Analyzer — rocket.zainrizvi.ca

- Created an LLM-powered analytics platform with **LangChain and Python**, using tool calling to query structured match metrics, player telemetry, and game outcomes
- Deployed a **Dockerized** 3D dashboard through **Cloudflare** to visualize gameplay events referenced by AI responses and synchronize them with match playback

Songify | AI Olympics Hackathon

- Awarded **1st place** at the AMACSS AI Olympics Hackathon with a live GenAI app using OpenAI API to create song-lyrics and album-art, demoed to **200+ attendees**